Calculus I

Tutorial #6 (04–08 Nov 2024)

Topics:

- Optimization problems
- L'Hospital's rule
- Newton's method
- Antiderivatives

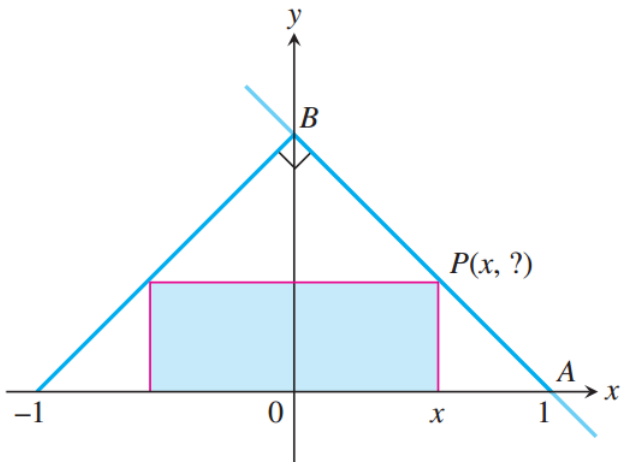
Problem 1. What is the smallest perimeter possible for a rectangle whose area 16 in^2? What are its dimensions?

Problem 2. The figure shows a rectangle inscribed in an isosceles right triangle whose hypotenuse is 2 units long.

a) Express the y-coordinate of P in terms of x. (Hint: Write an equation for the line AB.)

b) Express the area of the rectangle in terms of x.

c) What is the largest area the rectangle can have, and what are its dimensions?



Problem 3. You are planning to make an open rectangular box from an 8-in-by-15-in piece of cardboard by cutting congruent squares from the corners and folding up the sides. What are the dimensions of the box of largest volume you can make this way, and what is its volume?

Problem 4. *(The best fencing plan)* A rectangular plot of farmland will be bounded on one side by a river and on the other three sides by a single-strand electric fence. With 800m of wire at your disposal, what is the largest area you can enclose, and what are its dimensions?

Problem 5. *(Designing a tank)* Your iron works has contracted to design and build a $500 ft^3$, square-based, open-top, rectangular steel holding tank for a paper company. The tank is to be made by welding thin stainless steel plates together along their edges. As the production engineer, your job is to find dimensions for the base and height that will make the tank weigh as little as possible.

a) What dimensions do you tell the shop to use?

b) Briefly describe how you took weight into account.

Problem 6. Find the limits by using l'Hospital's rule

a) $\lim_{x\to 1} \dfrac{2x^2 - (3x+1)\sqrt{x} + 2}{x - 1}$

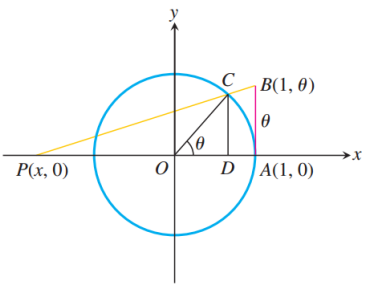
b) $\lim_{x\to 0} \dfrac{x(\cos x - 1)}{\sin x - x}$

Problem 7. Find a value of c that makes the function

$$f(x) = \begin{cases} \dfrac{9x - 3\sin 3x}{5x^3}, & \text{if} \quad x \neq 0, \\ c, & \text{if} \quad x = 0, \end{cases}$$

continuous $x = 0$.

Problem 8. In the accompanying figure, the circle has radius OA equal to 1, and AB is tangent to the circle at A. The arc AC has radian measure θ and the segment AB also has length θ. The line through B and C crosses the x-axis at P(x, 0).



a) Show that the length of PA is $1 - x = \dfrac{\theta(1 - \cos\theta)}{\theta - \sin\theta}$.

b) Find $\lim_{\theta \to 0}(1 - x)$.

c) Show that $\lim_{\theta \to \infty} \left[\dfrac{\theta(1 - \cos\theta)}{\theta - \sin\theta} - (1 - \cos\theta) \right] = 0$.

Problem 9.

a) Use Newton's method to estimate the two zeros of the function $f(x) = x^4 + x - 3$. Start with $x_0 = -1$ for the left-hand zero and with $x_0 = 1$ for the zero on the right. Then, in each case, find x_2.

b) Use Newton's method to find the positive fourth root of 2 by solving the equation $x^4 - 2 = 0$ Start with $x_0 = 1$ and find x_2.

Problem 10. *(Oscillation)* Show that if $h > 0$, applying Newton's method to

$$f(x) = \begin{cases} \sqrt{x}, & x \geq 0 \\ \sqrt{-x}, & x < 0 \end{cases}$$

leads to $x_1 = -h$ if $x_0 = h$ and to $x_1 = h$ if $x_0 = -h$. Draw a picture that shows what is going on.

Problem 11. Find the most general antiderivative or indefinite integral. Check your answers by differentiation.

a)
$$\int \frac{t\sqrt{t} + \sqrt{t}}{t^2} dt$$

b)
$$\int \frac{\csc\theta \cot\theta}{2} d\theta$$

THE END

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Answer to selected problems

1. 16 in^2
3. $V(5/3) = 2450/27$ in^3, $(5/3, 14/3, 35/3)$ in.
6. (a) -1, (b) 3
7. 27/10